

8	Answer: C
	$\phi = -\frac{GM}{r} = -\frac{GM}{(6.371 \times 10^6) + 80000} \dots\dots\dots (1)$ $g = -\frac{GM}{r^2} \Rightarrow 9.81 = -\frac{GM}{(6.371 \times 10^6)^2} \Rightarrow GM = 3.98 \times 10^{14} \dots\dots\dots (2)$ Sub (2) into (1): $\phi = -\frac{GM}{r} = -\frac{3.98 \times 10^{14}}{(6.371 \times 10^6) + 80000} = 62 \times 10^6 = 62 \text{ MJ}$

9	Answer: B
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